

## DBMS Lab 0

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Aim: To study SQL commands, their uses, syntax, and install MySQL for database management experiments.

### Q 1 What is SQL and its Use?

SQL is a standard language for accessing and manipulating databases.

SQL stands for Structured Query Language, which lets you access and manipulate data. It can perform CRUD operations i.e. Create, Read, Update and Delete the Data. One can create multiple tables with different data types in separate rows and columns. It can create various views in the database and can set permissions on the tables, procedures and views.

RDBMS (Relational Data Base Management System) is the basis of SQL and for all other modern database systems such as MS SQL Server, IBM DB2, Oracle, MySQL, and Microsoft Access.

### Q 2 Describe all the different commands of SQL, mention use of it and syntax.

SQL allows various actions which are acted upon via the use of SQL statements. SQL statements contain basic keywords and end with a semi colon ";".

Here are some examples of the keywords used in SQL :-

- SELECT - extracts data from a database

SELECT CustomerName, City FROM Customers

- UPDATE - updates data in a database

UPDATE table name SET column1 = value1, column2 = value2 WHERE condition;

- DELETE - deletes data from a database

DELETE FROM table name WHERE condition;

. **INSERT INTO** - inserts new data into a database

INSERT INTO table\_name VALUES (value1, value2, value3,  
... );

. **CREATE DATABASE** - creates a new database

CREATE DATABASE database\_name;

. **ALTER DATABASE** - modifies a database

ALTER DATABASE OldDatabaseName MODIFY NAME  
NewDatabaseName;

. **CREATE TABLE** - creates a new table

CREATE TABLE table\_name (column1 datatype constraint,  
column2 datatype constraint, column3 datatype constraint);

. **ALTER TABLE** - modifies a table

ALTER TABLE table\_name ADD column\_name datatype;

. **DROP TABLE** - deletes a table

DROP TABLE table\_name;

. **CREATE INDEX** - creates an index (search key)

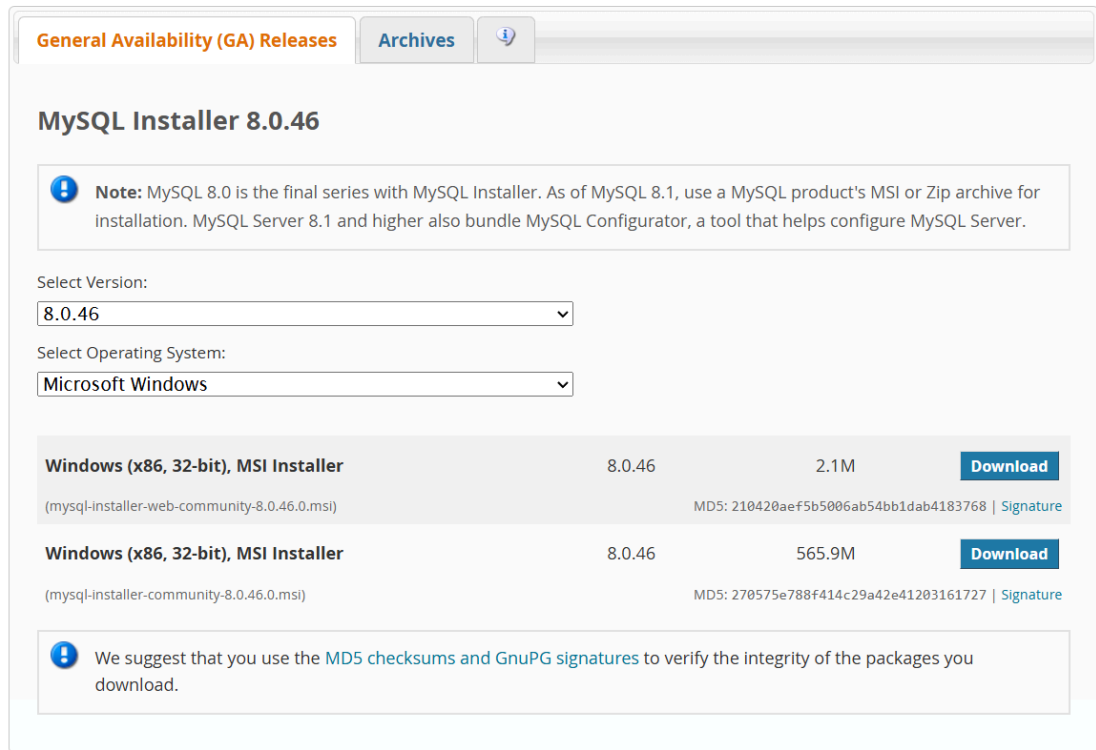
CREATE INDEX index\_name ON table\_name (column1, column2, ... );

. **DROP INDEX** - deletes an index

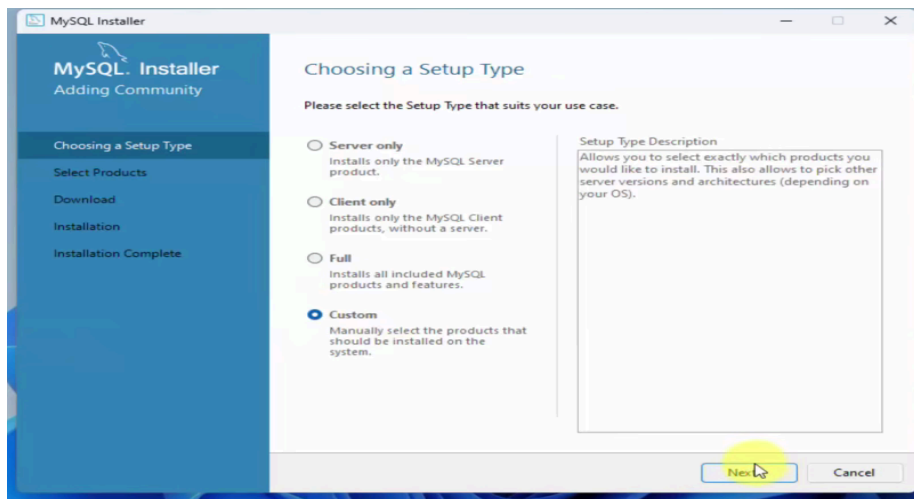
DROP INDEX table\_name.index\_name;

Q 3 Write a step by step procedure to install MYSQL.

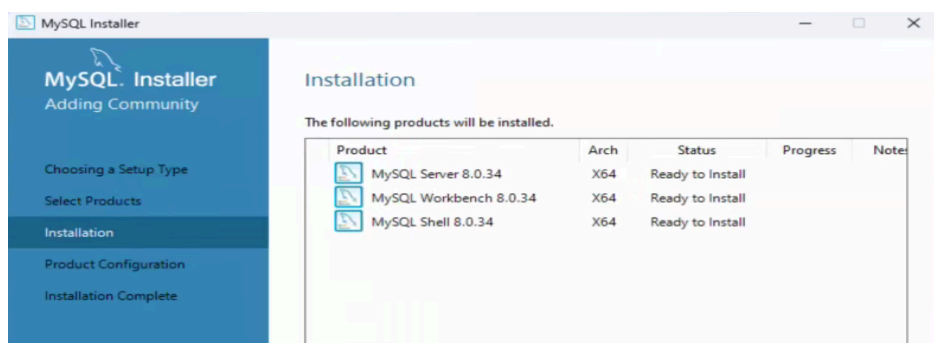
1. Download the MySQL Installer from the web ..



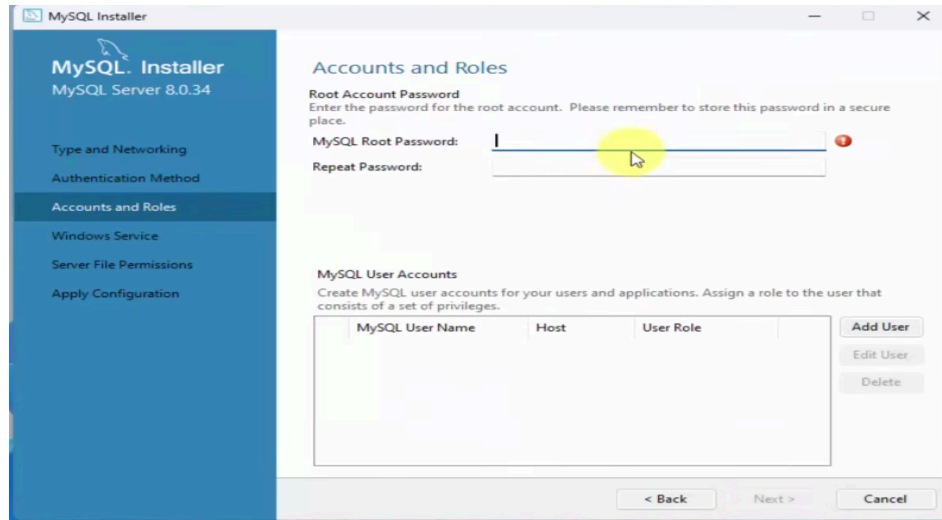
2. Run the installer and select the required setup type from the download window.



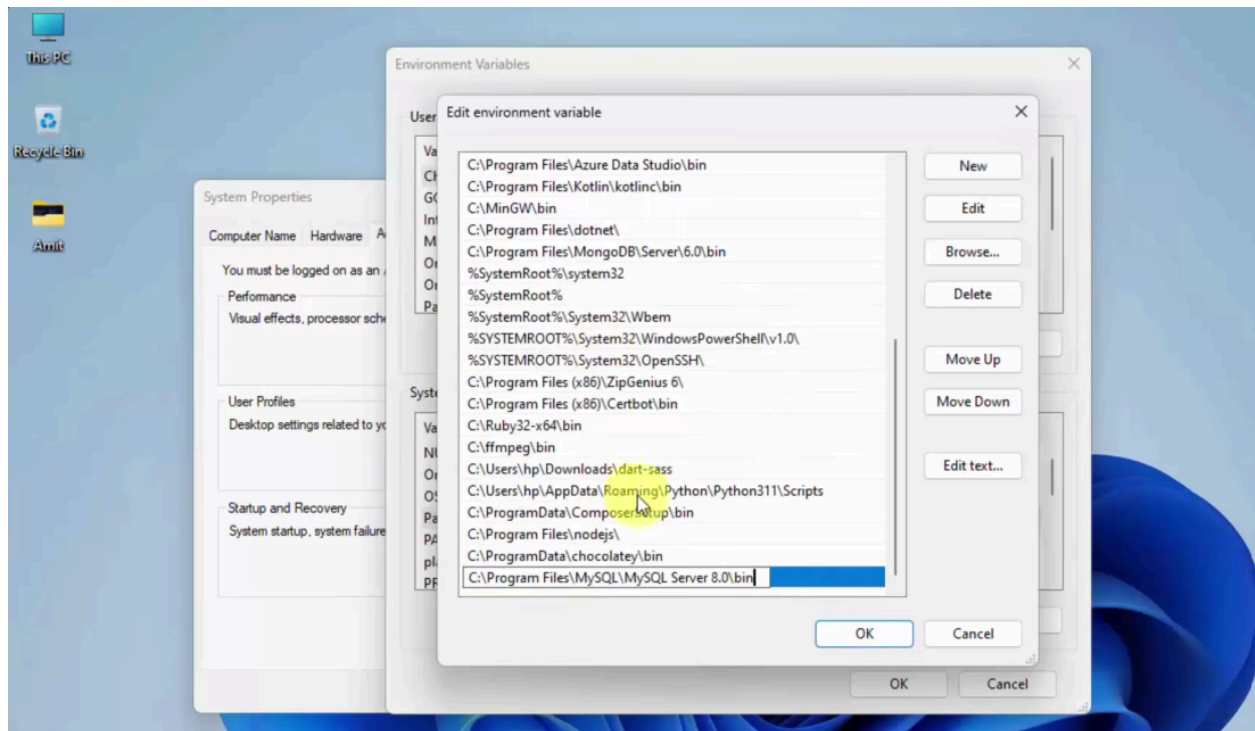
3. Install the required MySQL components.



#### 4. Configure MySQL Server.

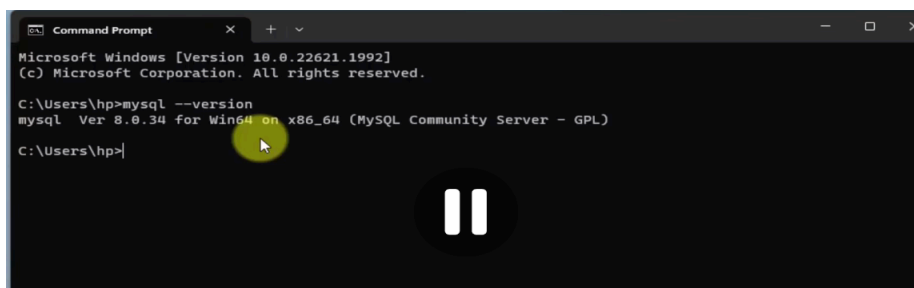


5. Set a password for the root user.



6. Complete the configuration process.

7. Open MySQL Workbench.



8. Connect using the MySQL credentials.

9. Verify installation using: `SHOW DATABASES;`

If databases are displayed, MySQL installation is successful.